Testing the Dorabella Cipher

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No cipher key tested turns the Dorabella into readable text. Two rounds of fair tests — about 30 in round 1 and 55 pre-registered test families in round 2 — all land at chance, while known text put through the same tests is recovered almost every time.

The cipher is not random: it has strong habits in which symbol follows which. But scrambled copies that keep those habits do as well as the real cipher in every test, in every language and style tried.

Round 2 at a glance (1–11 October)

Every round-2 family landed at chance against pair-preserving copies; the two closest calls, Latin and birthdays, fail the checks set in advance. Power is the share of planted messages that stood out from the copies.

Idea tested	Script	p vs pair-preserving copies	Power	Verdict
Transpositions built on 3, then any key (T3, TB)	three	0.35, 0.84	98%	Chance
Elgar's key through transpositions or period-3 shifts and turns (TE, PA, PB)	three	0.99, 0.46, 0.47	100%	Chance
Any key plus a period-3 shift or symbol turn (PC, PD)	three	0.41, 0.64	85%	Chance
Three alphabets used in turn (PE)	three	0.87	10%	Inconclusive: too much freedom
Loop counts alone, read in threes: repeats, counting keys, any key (L1, L2, L3)	looptrits	0.048, 0.26, 0.31	83%, 100%, 48%	Chance; any-key test low power

Idea tested	Script	p vs pair-preserving copies	Power	Verdict
63 cribs (greetings, names, places), any key	cribs	0.37	98%	Chance
Cribs with sets-of-three or loop-group keys	cribs	1.00, 1.00	97–100%	Chance
Quotation from Elgar's song and cantata texts (44,485 phrases)	quotes	0.75	98%	Chance
Quotation from Elgar's own letters (69,608 phrases)	quotes	0.48	97%	Chance
Quotation from the Psalms: Coverdale, King James, Douay	quotes	0.50, 0.47, 0.51	95–100%	Chance
Quotation from the 1662 Prayer Book, full KJV, full Douay	quotes	0.19, 0.66, 0.58	95–100%	Chance
Quotation from the texts appointed for 14 July 1897	quotes	0.73	95%	Chance
Playful spellings: no vowels, backwards, phonetic, initials, pig Latin	styles	0.26–0.88	92–100%	Chance
Elgar's private English: no AND, his abbreviations, his own letters	elgarstyle	0.26–0.36	99–100%	Chance
The date, July 14 97, as the key (Elgar's key; any key; transposition)	date	0.15, 0.38, 0.77	88–100%	Chance

Idea tested	Script	p vs pair-preserving copies	Power	Verdict
French, German, Italian, Latin	languages	0.14, 0.53, 0.075, 0.049	98–100%	Chance
Latin confirmation (Church Latin, pre-registered)	languages	0.049	100%	Not confirmed: decrypt unreadable
Musical scales as numbers	scales	0.41, 0.86, 0.72	90–100%	Chance
Birthdays, ages, line lengths 29-31-27	scales	0.63, 0.048, 0.12	90–100%	Chance: random numbers do as well
Note names CDEFGAB as the key	scales	0.66, 0.91, 0.41	95–100%	Chance
Every circle-and-cross key (589,824)	circlekeys	0.35	100%	Chance
Homophones: several symbols for one letter	homophones	0.37, 0.63, 0.30	62–78%	No evidence; low power

Where a row gives several p-values, they are the row's families in the order named. The line to beat in each script was 0.05 divided by its number of families, set before any run (0.0167 for three families, 0.00625 for the eight in three).

Round 1 results (27 September – 1 October)

The Dorabella beats plain shuffles of its own symbols, but never beats shuffles that keep its adjacent-pair habits, and never finds more real words than chance. Every one of 102 known-English controls scored far above it.

Idea tested	Script	Real Dorabella	Scrambled copies (mean / best)	p	Verdict
Any one-symbol-one-letter key, 160 reading configs, vs plain shuffles	marathon	z 5.28	2.21 / 5.68 (1,016)	0.0020	Order is not random

Idea tested	Script	Real Dorabella	Scrambled copies (mean / best)	p	Verdict
Same, vs pair-preserving shuffles	marathon	z 5.28	4.32 / 6.80 (1,016)	0.085	Explained by pair habits
Words in the best decrypt	marathon	11 words	p over 1,016 copies	0.35	Chance
Known English through the same pipeline	marathon	z 5.28	102 controls: median z 14.3, lowest 7.04	102 of 102 above	Not English-like
Key learned on 2 lines, scored on the 3rd, vs pair-preserving	marathon	-13.74	-14.00 / -12.50 (254)	0.26	Chance
Deeper search (simulated annealing), vs plain shuffles	marathon	-10.84	-11.37 / -11.00 (102)	0.0097	Order is not random
Loop count picks the letter group (1 loop = A-H etc.), direction the letter: all 24,576 keys x 160 configs, vs pair-preserving	loopgroups	-15.06	-15.14 / -14.54 (200)	0.30	Chance
Keypad-style keys: 287 triplet and affine layouts x 160 configs, vs plain shuffles	tests (Part C)	-16.27	-16.10 / -15.25 (300)	0.72	Chance
All 10.3 million sets-of-three keys x 160 configs (standard grouping)	allkeys	-13.84	-13.59 / -12.77 (50)	0.84	Chance
Same, alternative grouping (ABC DEF GHI KLM NOP QRS TUW XYZ)	allkeys	-14.05	-14.36 / -13.64 (50)	0.14	Chance
Keyword or phrase ordering the sets (261 phrase variants)	phrase / keyword	best p 0.003 (POWICK)	2 to 3 such hits expected by luck	—	Chance
TOMORROW as a Nihilist-style shift key	one-off	IC 0.0569	0.0567 (2,000)	0.44	Chance
Rare symbols are digits (Courage cards)	numbers	-10.80	-10.99 / -10.56 (100)	0.16	Chance

Idea tested	Script	Real Dorabella	Scrambled copies (mean / best)	p	Verdict
Rechecked transcription, 10 uncertain symbols solved in the search, English, vs pair-preserving	v2	-10.88	-10.93 / -10.41 (200)	0.33	Chance
Same, German (Fontane and two other 1890s novels)	v2	-10.98	-11.05 / -10.72 (200)	0.30	Chance

Scores are average log-probabilities of four-letter sequences; higher is more language-like. Real English of this length scores about -9.2, real German about -9.4. A p-value can never go below 1/(copies + 1).

The cipher as tested

Each of the 87 symbols is written as a two-digit code: loop count (1–3) then direction (1–8), with direction 1 = opening facing up. The common convention's direction 1 (opening to the right) is direction 3 here. Elgar's standard key, applied to these codes, reproduces the published decrypt (BLTACEIARWUNISNF…) at 80 of 87 positions; the misses fall on uncertain symbols.

```
line,codes
1,23 37 24 33 13 25 11 33 16 22 34 28 18 14 28 35 28 28 25 37 37 28 15 12 22
12 15 34 38
2,11 28 18 25 11 33 16 26 33 24 28 28 24 25 15 18 11 28 34 28 25 21 38 18 14
12 16 16 12 34 38
3,24 38 25 21 38 24 14 21 31 15 38 24 28 25 21 18 38 15 33 37 15 38 25 33 15
13 33
```

A careful recheck against a high-resolution image left 10 symbols uncertain. Every doubt is between two neighboring directions, never the loop count. A later look at the highest-resolution scan settled position 32 as 18, leaving 9. Under Elgar's key that is M, where the published decrypt has I.

Position	Main reading	Alternate
10	22	23

Position	Main reading	Alternate
23	15	14
24	12	11
25	22	23
32	18	settled (was 11)
34	11	12
67	14	15
78	15	14
85	15	14
86	13	12

The marathon, allkeys, numbers, phrase and keyword runs used the first transcription (five uncertain positions: 23, 32, 34, 78, 86; position 85 read as 14). The v2 runs used the rechecked one above, with all 1,024 combinations of the 10 readings open to the search (run before position 32 was settled).

How the tests work

Every test asks one question: does the real Dorabella do better than scrambled copies of itself put through exactly the same search? That corrects for trying millions of keys, which is where most proposed solutions fail.

1. **Score.** A decrypt is scored by how common its four-letter sequences are in period text: 2.5 million letters of 1890s English (Dracula), or 1.4 million of 1890s German.
2. **Search.** A hill climb (and, in the deep test, simulated annealing) looks for the key that maximizes that score, in each reading order: forward, fully reversed, each line reversed, and two alternating (boustrophedon) orders. Uncertain symbols are tried both ways.
3. **Scramble.** The same search is run on scrambled copies of the cipher. Plain shuffles put the 87 symbols in random order. Pair-preserving shuffles keep every adjacent symbol pair exactly as in the Dorabella, so they keep the writer's habits but carry no language.
4. **Compare.** The p-value is the share of copies that did as well as the real cipher. Beating plain shuffles only shows the order is not random; beating pair-preserving shuffles would point to something beyond habit.

5. **Check independently.** Word counts and held-out scoring (learn the key on two lines, score the third) are not what the search maximizes, so a real solution should pass them too.
6. **Prove the method works.** Known English (and German), enciphered with random keys and pushed through the same pipeline, sometimes with 3 deliberately misread symbols, shows what a genuine message looks like.

Findings by idea

Known English is easy to find; the Dorabella is not like it

All 102 English controls scored above the Dorabella (median z 14.3 against 5.28). Clean controls decoded at a median 99% of letters and those with 3 misread symbols at 90%; all 102 picked the right reading order. The Dorabella's best decrypt reads: OCTOSANDONAHUSERANBUTRAUSARAY / ILLITHAUSEYEWHNSREEROFLOWSHEWAY / NIGINEDDSEEMETHEYGLOWSCORDP.

The non-random order is writing habit

The Dorabella beat all but 1 of 1,016 plain shuffles, but 85 of 1,016 pair-preserving shuffles matched it. Most of its structure is which symbols tend to follow which, not language.

Loop count as the letter group

The first model tried: one loop for A-H, two for I/J-Q, three for R-Z (in any assignment), with direction picking the letter in any rotation, each group read either way. All 24,576 such keys in all 160 configurations give a best of -15.06, the same as plain shuffles (p = 0.40) and pair-preserving shuffles (p = 0.30), with fewer real words than either. The best decrypt reads INXTCTWEKKHEEMEWRENCYOSHUOAKB…

Elgar's alphabet in sets of three

Elgar's 1920 notebook groups letters in threes that alternate direction (A-B-C up, D-E-F down…). Every arrangement of the 8 sets around the 8 directions, each read up or down, is 8! x 2^8 = 10,321,920 keys; all were scored in all 160 reading configurations. Neither letter grouping beat chance; the standard grouping scored below the average scrambled copy.

A keyword Dora would know

No 8-letter word in 19,776 words of period vocabulary fits one letter per set. Treated as Victorian keyword alphabets, 64 phrases (DORABELLA, MISS PENNY, MISDORAPENY, WOLVERHAMPTON, ENIGMA, POWICK…) with no-vowel and initials variants gave only chance-level hits. TOMORROW, missing from the notebook's enciphered sentence, fails both as a keyword and as a Nihilist-style shift.

A second layer

A repeating shift of the Nihilist kind, which Elgar solved in 1896, flattens symbol frequencies (index of coincidence about 0.045). The Dorabella's is 0.058, inside the range of plain English (0.056 and up). Its frequencies look like a direct, one-layer cipher.

Numbers

The first Courage card shows ten loop symbols, as if mapping digits. Treating up to three of the six rarest symbols as digits changed nothing: best decrypt "qfsd#ildu#me THE BEE…", p = 0.16, fewer words than the copies. Planted English with a date and a train time was recovered at 99%.

The rechecked transcription and German

With the 10 uncertain symbols solved inside the search, English and German both land in the middle of the pair-preserving copies (p = 0.33 and 0.30). Planted English and German notes were recovered at 97% and 98%.

Findings, round 2

Stricter rules for round 2

Round 1 showed how easily one lucky run can look like a signal, so round 2 tightened the method.

- **Pre-registered families.** Each script's ideas, keys and texts were written into the script before any real run, and never changed after a result came in.
- **A corrected line to beat.** With several families in one script, a result counts only below 0.05 divided by the number of families (Bonferroni).
- **The real cipher's score is a median.** It is taken over 5 to 9 search seeds, not the luckiest seed (the lesson of round 1's held-out check).

- **Random-key comparisons.** Where a test used meaningful numbers (a date, birthdays, scales), the real cipher was also run with random numbers of the same size. Special numbers must beat random ones.
- **Uncertain symbols inside the search.** The nine uncertain symbols were chosen by the search wherever the script allowed it.
- **Power measured every time.** Planted text, written the way each test assumes, sometimes with 2 misread symbols, shows how often a real message would stand out.
- **Confirmation runs.** Anything near the line got a fresh, pre-registered run on new data, with the rule for success written down first.

Built on three

Elgar's alphabet is built on 3 (one, two or three loops) and the message is 87 = 3 × 29 symbols, so eight families tested schemes built on 3. None beat the pair-preserving copies; the line to beat was 0.00625.

- **Transpositions, then any key.** 210 grid, column, rail-fence and every-third-symbol orders built on 3 ($p = 0.35$), and 546 broader ones such as columnar widths 2–30 ($p = 0.84$).
- **Elgar's own key through those orders, or with a shift or symbol turn that repeats every 3 symbols** (all 24^3 shifts and 48^3 turns): $p = 0.99$, 0.46 and 0.47.
- **Any key plus a repeating shift or turn:** $p = 0.41$ and 0.64, power 85%.
- **Three separate alphabets used in turn:** $p = 0.87$, but with three free keys on 87 symbols, planted English stood out only 10% of the time. That family is inconclusive, not a refutation.

One side measure came close: the word count of the best grid-order decrypt ($p = 0.018$). It is not the pre-registered statistic, and its decrypt reads "…OTHER WILL … THEN WHY … NOT … SAD SET AS…", with no sentence.

Loop counts alone. Three loop counts make a base-3 number with 27 values, enough for 26 letters and a space, so the 87 symbols could carry a 29-character message in their loops alone. Three tests, line to beat 0.0167:

- **Do the 29 triples repeat like letters?** A little: 17 different triples, with 232 five times (index of coincidence 0.052, $p = 0.048$ against 2,000 pair-preserving copies, power 83%). It misses the line.

- **Counting keys** (triple read as a number, any rotation of A–Z plus space, 5 reading orders): p = 0.26, power 100%. The best reading is SVISDYTLL_MISPLHPUCKSSHJPI_ZE.
- **Any key:** p = 0.31, but with only 29 letters planted English stood out just 48% of the time, so this one is inconclusive. Its best reading, "…WHIMS OF YOUR ME THEN G AND ARE COMM", shows how easily 29 free letters make words.

Cribs

A crib is a word the writer probably used. Fixing one in place and searching for the rest found nothing beyond chance (p = 0.37 against 1,289 pair-preserving copies, power 98%).

- **The list.** 63 cribs fixed before any run: greetings and sign-offs, Penny and Elgar family names, Wolverhampton places, friends from the Variations, and the notebook sentence DO YOU GO TO LONDON TOMORROW. Short words such as DORA or ELGAR were left out by a rule decided on scrambled copies only: they fit almost anywhere.
- **Best fit:** GRIFFITH, in a decrypt reading "…TELL NEE MET HER … BADLY GRIF FIT H GONE … DEED AS FAIN HE WAIT…".
- **Sets-of-three and loop-group keys** (the two spreadsheet approaches: sorting by direction and by loop count): no such key spells any eligible crib anywhere in the real cipher.

MY DEAR DORA. DEAR DORA fits the start of line 3, read backwards, and only 41 of 1,289 copies had any such fit. That looks striking (p = 0.0047), but it was noticed by looking at the cipher, so it was reported as exploratory from the start. And the text completed around it is worse than the ordinary best fit (-12.08 against -11.33): "…DON ARE Y DEAR DORA RGLIILBUR DON GN HUE…". A real greeting would bring readable text with it.

Quotations

If the note quoted a text Elgar knew, a phrase from it should fit the cipher and the rest should then decode. It never did: across 8.0 million phrases from nine sources, the real cipher matched its copies every time.

The test places each phrase (12–30 letters; 14–30 for the largest sources) wherever its pattern of repeated letters fits the cipher, completes the rest with the best key, and keeps the best result. A second measure counts how many long phrases fit at all.

Source	Phrases	p, best completion	Long phrases fitting: real vs copies' mean
Words Elgar set: 18 songs and cantatas	44,485	0.75	4 vs 5.2
Elgar's own letters to 1897	69,608	0.48	5 vs 7.4
Psalms, Coverdale (Prayer Book Psalter)	204,464	0.50	21 vs 21.9
Psalms, King James	187,943	0.47	18 vs 20.4
Psalms, Douay	232,056	0.51	27 vs 22.7
Book of Common Prayer, 1662	647,520	0.19	60 vs 70.1
King James Bible, complete	2,914,078	0.66	265 vs 250.4
Douay Bible, complete	3,699,977	0.58	395 vs 329.2
Texts appointed for 14 July 1897	31,776	0.73	5 vs 3.3

The 14 July 1897 set is the Psalms for day 14 (71–74), Psalms 7, 14, 97 and 98 (from the date's numbers), the lessons in both the 1871 and 1662 tables, and the collect, epistle and gospel for the fourth Sunday after Trinity. With millions of phrases, some always fit: the best real completions were "FOR THE SINGER" (from the letters), "A WORLD OF CARE" (The Banner of St George) and "OF YOUR WIVES AND" (Douay), each surrounded by noise. Planted messages containing a quotation stood out 95–100% of the time.

Writing styles

Round 1 ended by asking whether Elgar wrote the note in a playful or private style that a standard English scorer cannot reward. Round 2 built a scorer for each style; none helped.

Each style's model was trained on Dracula rewritten that way, and planted text was written the same way. The search then ran exactly as for plain English.

Style	p vs pair-preserving copies	Power
Standard spelling (reference)	0.35	100%

Style	p vs pair-preserving copies	Power
Vowels dropped	0.88	98%
Words spelled backwards	0.26	96%
Phonetic spelling	0.62	100%
First letters of words only	0.66	92%
Pig Latin	0.33	100%
No AND (Elgar's "&" left unwritten)	0.26	100%
Elgar's shorthand (wd, shd, Xmas, thro', Septr…)	0.36	99%
Elgar's own voice (his letters added to the model)	0.30	99%

The first six used 2,000 copies each (line to beat 0.0083); the last three, 1,000 (line 0.0167). Word counts were at chance too.

Elgar's private English. The last three styles come from his own letters before 14 July 1897, transcribed from J. N. Moore's *Edward Elgar: Letters of a Lifetime* (about 13,300 words, pages 1–59). In private he wrote "&" 429 times and "and" only 50, yet his one newspaper article has 36 "and"s and no "&". His cipher alphabet has no symbol for "&", so a note in his private style might simply lack its ANDs. It made no difference. The transcription stays private for copyright reasons; only these counts are shared.

The date as the key

The note is dated 14 July 1897, so its date may be the key. Seventeen forms of the date, used three ways, did no better than chance or than random numbers of the same size.

- **The keys.** Every common written form: 71497, 7141897, 071497, 14797, 140797, 970714, 18970714, 714, 147, the day of the year (195) and its year forms, and the numbers 7, 14, 97 in both orders.
- **The uses.** A letter shift, a symbol turn (in 45-degree steps) or a loop-count step, forwards or backwards, running through the message or restarting each line; digit keys also as column orders for a transposition.

Family	p vs pair-preserving copies	p vs random numbers	Power
Elgar's own key with the date settings (no search)	0.15	0.077	100%
Any key plus a date shift, turn or step	0.38	0.25	88%
Any key plus a date transposition	0.77	0.64	95%

Because any key is allowed, a constant added to every shift is absorbed by the key, so the starting value of a date never matters in the last two families.

Other languages

No language reads. Latin came closest twice (p = 0.049 both times), but a pre-registered confirmation failed the test of readability, so Latin is not confirmed.

Language	Model text	p vs pair-preserving copies	Power
English (reference)	Dracula	0.16	97%
French	Hugo, *Les Misérables*, tome I	0.14	100%
German	Fontane, *Effi Briest*; Goethe, *Werther*	0.53	100%
Italian	Manzoni, *I promessi sposi*	0.075	100%
Latin	Vulgate Daniel; Caesar; Cicero; Horace	0.049	98%
Latin, confirmation	Vulgate Psalms and Gospels (Church Latin)	0.049	100%

Each model learned from the first 85% of its text; planted messages came from the held-back 15%. The line to beat for the four new languages was 0.0125.

The Latin confirmation. The rule, written into the script before the run: Latin is confirmed only if the cipher beats the pair-preserving copies at $p < 0.05$ *and* the decrypt reads as Latin. It met the first by a hair (48 of 1,000 copies did as well) and failed the second. The

best decrypt reads "AORAUIC FACIS QUETICHQCTI QUI TIMOLLORSI QUEM … ENIM … ORE … DAT", while planted Latin came back 95–99% right. The cipher scores -11.05, its copies -11.32 on average, and real Latin about -9.1: a slight lean, nowhere near a message. The searches keep reaching for QUI, QUE and -US, which a few repeated symbol pairs can mimic.

Music and numbers

A composer writing to a musical young woman might key a cipher to a scale, and Elgar's 8 directions match the 8 notes of an octave. Three sets of musical and personal numbers, each used the same three ways as the date, gave nothing.

- **Scales as numbers:** the step patterns of the major, three minor and four other modes (2212221 and so on), the notes of three scales in semitones, and the octave climbing and falling.
- **Note names as the key:** A to G as 0 to 6, every scale written as letters (C major = C D E F G A B), rising and falling: 14 orders.
- **Personal numbers:** the birthdays of Elgar (2 June 1857), Dora (8 February 1874), Alice (9 October 1848) and Carice (14 August 1890) in several forms; their ages in July 1897 (40 and 23); and the line lengths 29, 31, 27.

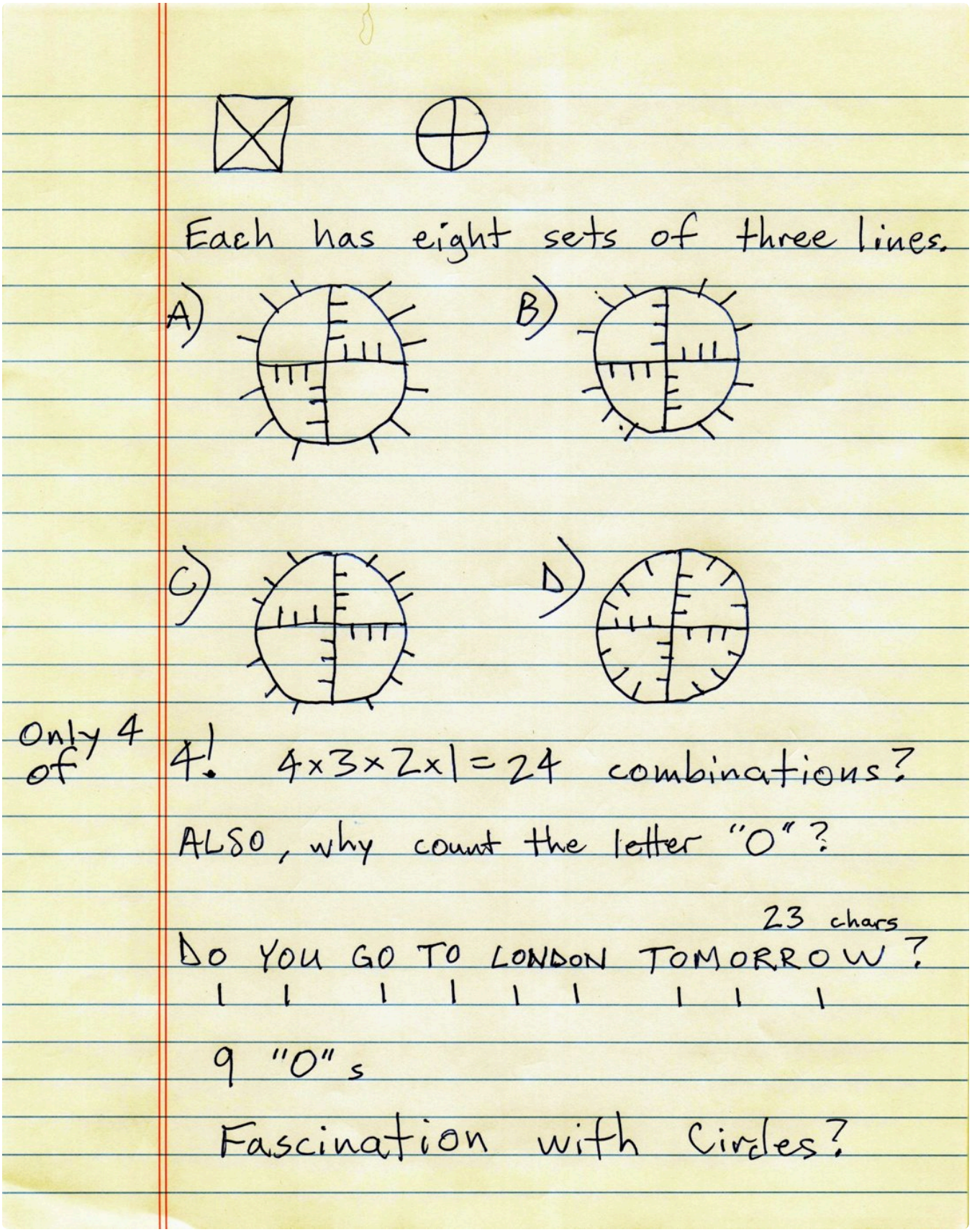
Key set	Elgar's key (no search)	Any key + shift, turn or step	Any key + transposition
Scales as numbers	0.41	0.86	0.72
Note names	0.66	0.91	0.41
Birthdays, ages, line lengths	0.63	0.048	0.12

Values are p against pair-preserving copies; power was 90–100% throughout, and the line to beat 0.0167.

The one value under 0.05, birthdays with any key, fails twice: it misses the line, and random numbers of the same size did as well (p = 0.35). Its decrypt reads "R PORCH URBLIN NON HT TEMPTS FLY SID AND HARK … TIES … THAT … DEAD". The note-name orders are rotations of one pattern, so they scored a little below random orders (p = 1.00 against random keys); that can only make them look less special.

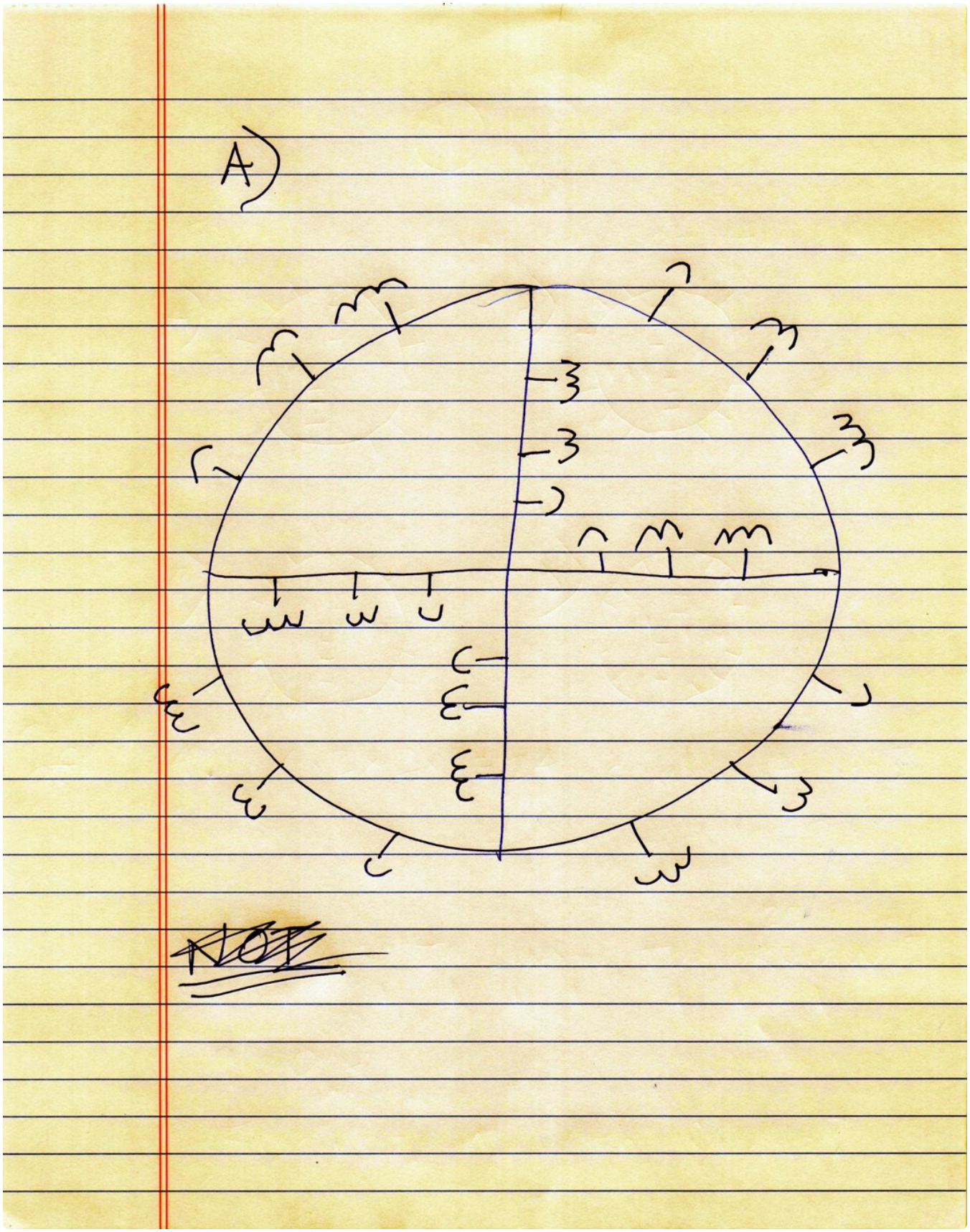
The circle and the cross

Elgar's 1920 notebook page shows circles divided by a cross. Redrawn at a larger scale, it suggests a layout for his 24 symbols different from the usual eight spokes — and his own key follows it. No key drawn this way reads, though (p = 0.35, power 100%).

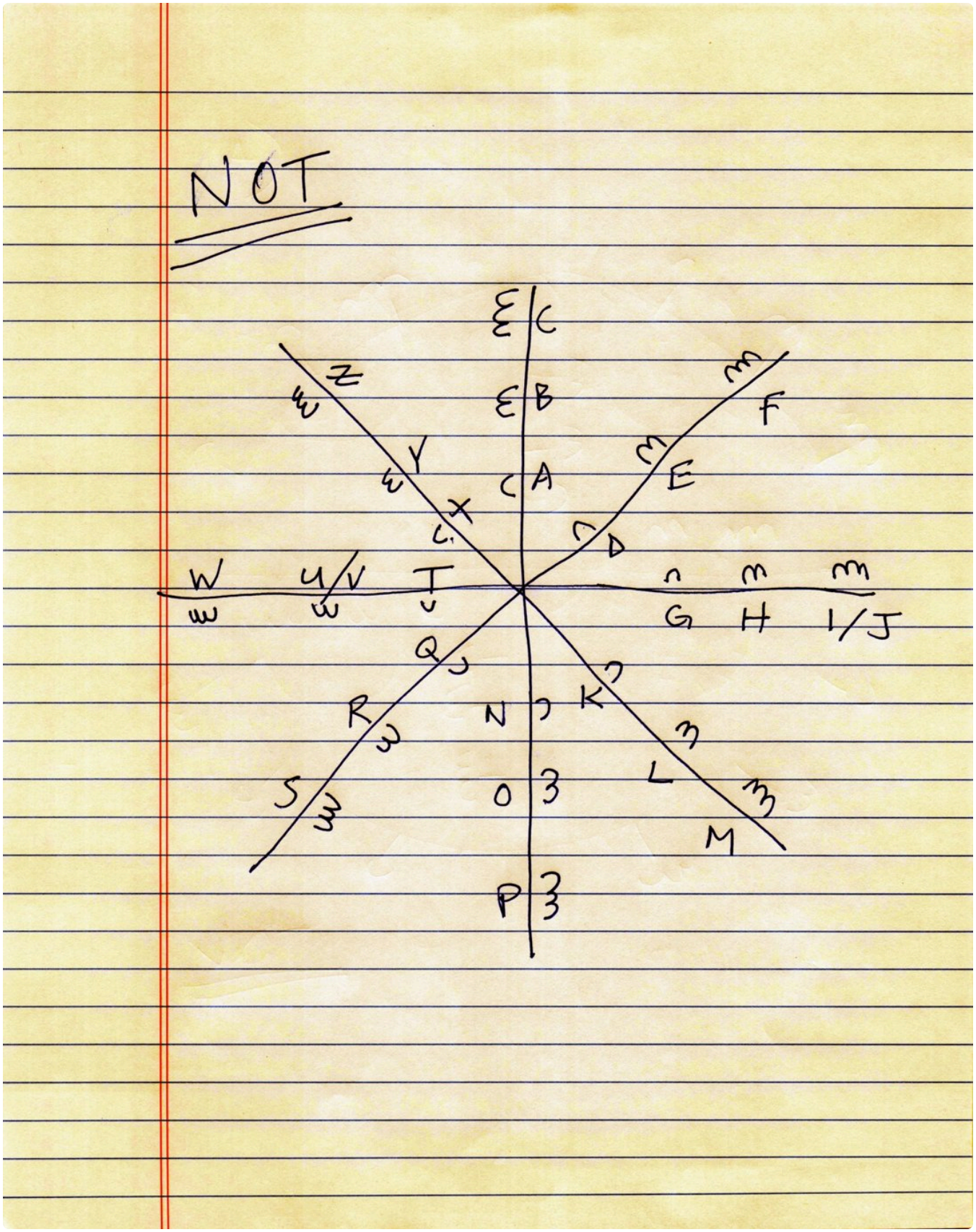


The investigator's redrawing of the notebook page: "each has eight sets of three lines".

On each circle the four arms carry three ticks and each quarter of the rim carries three strokes: eight sets of three. Drawn with Elgar's loops, the arm ticks face up, down, left and right, and the rim strokes face the four directions in between. That gives exactly his 24 symbols, with no spokes needed.



Circle A drawn with the cipher's own symbols.



The usual eight-spoke picture, which the redrawing argues against.

Elgar's key fills one cross, then the other. With the directions numbered 1–8 in 45-degree steps, his published key puts ABC, DEF, GHI and JKL on directions 3, 5, 1 and 7, four positions 90 degrees apart. MNO, PQR, STU and WXY go on 8, 6, 4 and 2, the other cross. That is the arms-then-rim layout of the notebook circles, not a walk round the spokes.

Every such picture tested. The test tried every key the figure allows, with no search: either letter grouping; either cross first; all 24 × 24 orders of the groups round the two crosses (every choice of which side the ticks sit and whether rim strokes point in or out is one of these); each group of three running up or down. That is 589,824 keys, each read in the 5 reading orders, against 2,000 pair-preserving copies. Elgar's own key is one of them, and it scores -15.21, worse than the best of almost any scrambled copy. These keys are a subset of round 1's 10.3 million sets-of-three keys, now tested against the stronger, pair-preserving copies.

Homophones: where this method runs out

Giving common letters several symbols found nothing, but this is the one test too weak to rule its idea out.

Why it was tried. On the same notebook page, as the investigator reads it, Elgar wrote out DO YOU GO TO LONDON TOMORROW, ticked every O, and noted "9 O's" and "23": nine of the 23 letters are one letter (see the bottom of the first drawing above). Someone counting a common letter is looking at the weakness of a one-symbol-one-letter cipher, and the classic cure is homophones: several symbols for E, O or T. Every earlier search allowed only one letter per symbol.

Family	p vs pair-preserving copies	Power	Planted letters decoded right (median)
Up to 2 symbols per letter	0.37	62%	38%
Up to 3 symbols per letter	0.63	78%	38%
Extra symbols for common letters only (ETAO up to 3, INSHR up to 2)	0.30	65%	10%

The decrypts are packed with THE, THAT and THERE ("…THAT THREE … THAT O HAUE TO THAT…"), the mark of too much freedom. Planted messages often scored *below* some other reading the search found, so they were recovered badly. With only 87 symbols,

many homophonic keys produce equally English-looking text; the message is shorter than its unicity distance. Longer searches made this worse, not better.

Verdict: no evidence for homophones, but not excluded. A homophonic Dorabella would have been flagged only about two times in three, and rarely read correctly.

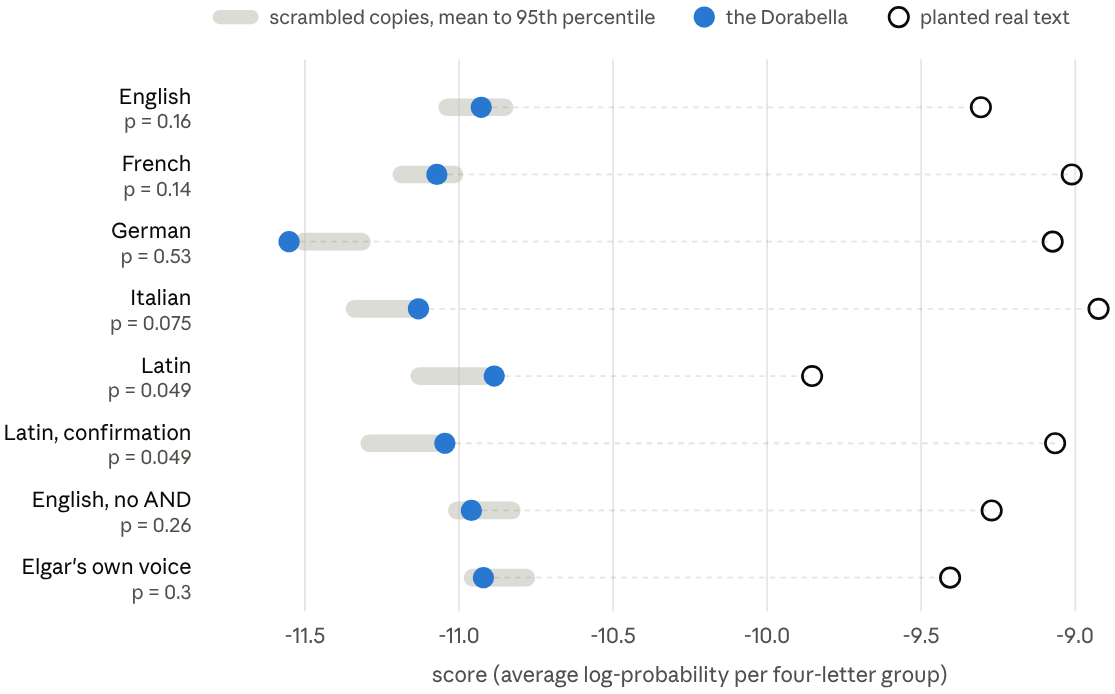
What the pattern says

In every language and style, the Dorabella scores at or a little above the middle of its own scrambled copies, and nowhere near real text.

The cipher sits with its scrambled copies, far from real text

Average lift over the copies' mean: 0.15. Average gap to planted real text: 1.82.

Average four-letter score in each row's own language model; higher reads more like the language.



Final reports: languages_out, latin_confirm_out, elgarstyle_out · 1,000 pair-preserving copies per row; planted text with no misreads

The cipher clearly beats plain shuffles (p = 0.001 to 0.009 in seven of these eight rows) but not copies that keep its adjacent symbol pairs. So it has real, measurable structure in which symbol follows which, and language models reward that structure a little. That is the honest core of the impression that "it is a language". What no test found is a message on top of that structure: the gap to real text is 1.0 to 2.5 points in every row, far more than the cipher's lift over its copies.

Open threads

Two small signals survived round 1, none strong enough to call a result. A third did not survive a recheck.

- **Position 32 is 18.** The search favored 18 over 11 more than chance allows (p = 0.0088 against pair-preserving shuffles, 8 of 1,016), and both v2 searches chose it. The highest-resolution scan then confirmed 18. That recheck was made knowing what the statistics favored, so a blind second reading would make it airtight. Position 67 as 15 is the next reading to check.
- **Reversed reading orders.** Fully reversed and each-line-reversed beat pair-preserving shuffles modestly (p = 0.018 and 0.021, 1,016 copies). About ten such order and reading comparisons were made, so one or two at this level is expected by chance.
- **A held-out result that did not hold up.** One pre-registered configuration (each line reversed, with 32=18, 34=12, 78=14, 86=12) hit the floor on the held-out test against both kinds of copy (p = 0.0039, 254 copies). But the held-out score depends on the random seed of the climb. Rerun with 24 seeds, the same configuration scored -16.5 to -13.8 (median -15.05); the marathon's single run had drawn the best. At the typical score, p = 0.20 against 240 pair-preserving copies, and the held-out lines decode to gibberish. Lesson: compare a seed-dependent statistic using the real cipher's typical value over several seeds.

After round 1, the largest remaining unknown was the plaintext, not the key. Elgar's own key gives English-like letter frequencies but not English-like order, which fits phonetic spelling, private abbreviations or in-jokes — none of which a standard English scorer can reward. Round 2 built scorers for those styles, including Elgar's own private English; none helped (Writing styles, above).

After round 2. Five results came near 0.05; each is explained, and none reads. The fifth is the loop-count repeats (0.048), which miss the line and decode to nothing under any counting key.

- **Latin, 0.049 twice.** The pre-registered confirmation failed its readability test. Treat it as the language-like pair structure shown in the chart above, not a Latin message.
- **Birthdays with any key, 0.048.** It misses the corrected line, and random numbers do as well (p = 0.35).
- **Italian, 0.075.** Not below 0.05.
- **MY DEAR DORA, 0.0047.** Found by looking, so exploratory by design; the completed text does not read.

Across more than 80 tests, a few results near 0.05 are expected by chance.

Three things this method cannot settle:

- **Homophones.** At 87 symbols, a cipher with several symbols per letter cannot be reliably read by statistical search (power 62–78%).
- **Three alphabets in turn.** The same problem: power 10%.
- **The transcription.** Nine symbols remain uncertain. A blind re-reading by someone who has not seen these results would make positions 32 and 67 airtight.

Ideas not yet tested: names of God and Jesus as cribs, street names and addresses as cribs, and the cipher's single dot as "&".

Reproducing this

All scripts, logs and the transcription are in the GitHub repository cjager13/Dorabella-cipher-tests. They are plain Python 3 and run on an 8-core laptop; allkeys also needs numpy. Keep Dorabella_hillclimb.py, Dorabella_tests.py and english.txt in the same folder, since the others load them. Every script has a --selftest or planted-text check; run it first.

Script	What it tests	Time (8-core laptop)
Dorabella_marathon.py	Main search, both scramble types, known-English controls, held-out, deep search; resumable, runs until stopped	18.6 h first pass; 46.6 h for 1,016 copies
Dorabella_allkeys.py --all-configs --nulls 50	All 10,321,920 sets-of-three keys x 160 configs	16.8 h
Dorabella_numbers.py	Rare symbols as digits	1.6 h
Dorabella_v2.py (--lang de for German)	Rechecked transcription, uncertain symbols solved in the search	45 min each
Dorabella_verify.py	160-config search vs 20 identical searches on shuffles	20 min
Dorabella_phrase.py	Keyword and phrase alphabets	30 s

Script	What it tests	Time (8-core laptop)
Dorabella_keyword.py	8-letter keywords, one letter per set	seconds
Dorabella_loopgroups.py	All 24,576 loop-group keys vs both kinds of shuffle (needs numpy)	about 10 min
Dorabella_heldout_check.py	Reruns one held-out configuration over many seeds and copies (the correction above)	about 25 min

The German corpus is spelled out without umlauts (ä to AE, ö to OE, ü to UE, ß to SS) and stripped of Project Gutenberg's English license text.

The first day's outputs (27 September 2026) are in the repository's early_runs folder, summarized in its README. One of them matters for everything above: planted English with 2, 4 or 8 misread symbols was recovered at only 87%, 48% and 30% by the early solver, so a few transcription errors could hide a message. That is why the transcription was rechecked and its uncertain symbols built into the v2 search.

Round 2 scripts. Round 2 took about 240 hours of laptop time. Every script resumes after Ctrl+C, has a --selftest, and with --stop-when-done ends when its pre-set number of copies is reached.

Script	What it tests	Time (8-core laptop)
Dorabella_three.py	Transpositions and period-3 keys (8 families)	78 h
Dorabella_cribs.py	63 cribs, three key types, MY DEAR DORA	18.5 h
Dorabella_quotes.py --source <folder>	Quotation from one folder of texts	2.7–11.9 h per source; 65 h for all nine
Dorabella_styles.py	Five playful spellings and standard English	18.5 h

Script	What it tests	Time (8-core laptop)
Dorabella_elgarstyle.py	Elgar's private English (needs the private letters file for its last style)	8.3 h
Dorabella_date.py	The date as the key	8.3 h
Dorabella_languages.py (--confirm-latin)	French, German, Italian, Latin; the Latin confirmation	12.5 h; 1.5 h
Dorabella_scales.py --set scales, numbers or notes	Music and personal numbers as the key	6–7.5 h each
Dorabella_circlekeys.py	Every circle-and-cross key	38 min
Dorabella_homophones.py	Several symbols per letter	3.1 h
Dorabella_looptrits.py --workers 8	Loop counts alone, read three at a time (not resumable)	6 min
Dorabella_fixtext.py, Dorabella_pdf2txt.py, Dorabella_daytexts.py	Helpers: repair PDF and OCR text; build the 14 July 1897 texts	seconds

The Bible and Prayer Book texts were cleaned before use: ligatures and long-s splits repaired, small-capital OCR errors fixed ("Lorp" to Lord), and the Coverdale Psalms' Latin titles kept. The King James text is Project Gutenberg's eBook #10, and the Church Latin text is the Clementine Vulgate from eBible.org. All of round 2 was scored against the same english.txt (Dracula, Gutenberg #345).

Private material. The transcription of Elgar's letters from Moore's *Letters of a Lifetime* is not in the repository, for copyright reasons. Only derived counts and the short phrases quoted here are shared. Without that file, Dorabella_elgarstyle.py runs its first three styles; the letters quotation test cannot run.

Sources

- Cipher Mysteries: The Dorabella cipher and Elgar's other little ciphertexts — Courage cards, the 1896 Nihilist cipher, the 1920 notebook
- Cipher Mysteries: Dorabella Cipher — the standard decrypt
- NumberWorld: The Dorabella Cipher (Part 3)

- Wikipedia: Dorabella Cipher
- GitHub: cjager13/Dorabella-cipher-tests — scripts, logs and transcription
- Corpora: Bram Stoker, *Dracula* (1897), and Theodor Fontane, *Effi Briest* (1895), plus two other 1890s German novels, all from Project Gutenberg
- Jerrold Northrop Moore, *Edward Elgar: Letters of a Lifetime* (Oxford, 1990) — Elgar's letters to July 1897, used privately for the writing-style and letters tests
- Project Gutenberg eBook #10: The King James Bible
- eBible.org: Latin Clementine Vulgate (latVUC) — the Church Latin confirmation text
- Round 2 corpora, all public domain: Hugo, *Les Misérables* tome I (French); Goethe, *Die Leiden des jungen Werthers* (German); Manzoni, *I promessi sposi* (Italian); the Vulgate's Daniel, Caesar, Cicero and Horace (Latin); the Douay-Rheims Bible; and the 1662 Book of Common Prayer from a scanned PDF